

Data Collection Report

Project: AI Football League Table Predictor (End-of-Season)

1. How did you obtain your data, if it's a Data-driven problem?

This is a data-driven machine learning problem aimed at predicting the final end-of-season football league tables. The data was obtained dynamically via a Python script interfacing with the Understat platform, a leading resource for advanced football analytics. The dataset comprises historical match data for the Top 5 European leagues (English Premier League, La Liga, Serie A, Bundesliga, Ligue 1) from the 2014 to 2024 seasons. It includes match results, team identifiers, and advanced metrics—most notably Expected Goals (xG) for both home and away teams, alongside pre-calculated match outcome forecasts.

2. How large is your data?

The dataset is comprehensive, containing exactly 18,085 matches played by 161 unique teams across 10 seasons. Each match record initially contains 13 raw statistical features. After feature engineering (calculating rolling averages, Elo ratings, and rest days), the feature space expands significantly. To ensure robust model evaluation and prevent temporal leakage, the data is split chronologically into Training (2014–2021), Validation (2022), Testing (2023 season)

3. In what format are you storing your data or states? Describe the abstract data type, not just the fileformat.

Fundamentally, the dataset is modeled and stored as a **2D Feature Matrix** ($X \in \mathbb{R}^{n \times d}$). In this abstract data type, each row represents a single match sample, and each column represents an engineered feature (e.g., home team rolling xG, away team Elo rating). This matrix representation is the native and optimal input structure for our chosen XGBoost gradient boosting classifier. Additionally, during the feature engineering phase, the data is conceptually treated as a **Directed Weighted Graph** (where nodes are teams, directed edges are matches, and weights are xG differences) to dynamically compute and update team Elo ratings over time.

4. Did you need to process the original data to get it into an easier, more compressed format?

Yes. The original data was retrieved as highly nested JSON objects. This was flattened and converted into a structured CSV format. Significant processing was required to engineer temporal features; we computed prematch rolling averages (e.g., last 5 matches' xG and points) using strictly historical data to avoid look-ahead bias. Finally, categorical variables (team names) were encoded, and numerical features were scaled based on training set statistics to finalize the 2D matrix for the XGBoost model.

5. How would you simulate similar data or your game?

To simulate similar data and generate the final league tables, we employ a **Monte Carlo simulation** framework. The trained XGBoost model acts as our probabilistic engine, outputting a probability distribution (Home Win, Draw, Away Win) for any given fixture based on the teams' current feature states. To simulate a season, we feed the remaining unplayed fixtures into the model, sample match outcomes from the predicted probability distributions, and update a synthetic league table (points and goal differences). By repeating this stochastic simulation 10,000 to 50,000 times, we generate a robust statistical distribution of final standings, allowing us to extract the probability of each team winning the title, qualifying for European tournaments, or facing relegation.